CURRICULUM VITAE

Saeed Rafieyan

Also listed as Saeed Rafieyan Boroujeni

PhD researcher, Digital Manufacturing of Biomimetic Systems group, Institute of Physical Chemistry, Polish Academy of Sciences, Warsaw
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EDUCATION

2026 – present

PhD

Institute of Physical Chemistry, Polish Academy of Sciences (IChF PAN) · Warsaw, Poland

- Digital Manufacturing of Biomimetic Systems group.
- Research topic: computational inverse design for biodegradable wound dressings.

2018 – 2022

MSc in Chemical Engineering – Biomedical Engineering

Tarbiat Modares University · Tehran, Iran

Thesis: *Predicting cell behavior on cardiac tissue engineering scaffolds using machine learning algorithms*

- GPA: 3.78 / 4

2013 – 2017

BSc in Chemical Engineering

Tafresh University · Tafresh, Iran

Thesis: *Simulation of the biodiesel production process with Aspen HYSYS*

RESEARCH EXPERIENCE

2026 – present

PhD researcher – Digital Manufacturing of Biomimetic Systems group

Institute of Physical Chemistry, Polish Academy of Sciences (IChF PAN) · Warsaw, Poland

- Developing computational approaches that connect design requirements with candidate scaffold structures, with applications in regenerative medicine and wound healing.

2025 – present

Research assistant – data scientist

Jiangnan University, Dr Madadi's group · Wuxi, China

- Machine-learning and deep-learning models for biofuel production and process optimisation, contributing to a peer-reviewed article in Bioresource Technology (2025).

2023 – 2025

Research assistant – data scientist

Tarbiat Modares University, Dr Fakhroleslam's group · Tehran, Iran

- Machine-learning and deep-learning models for distillation and separation processes; chemical-engineering text mining.

2019 – 2025

Research assistant – data scientist

Tarbiat Modares University, Prof Vasheghani-Farahani's group · Tehran, Iran

- Led the development of MLATE through three published versions as first author, including its literature-derived datasets and web app (mlate.ir).
- Developed TissueGPT, a language model adapted to tissue-engineering literature, released on Hugging Face.

2022 – 2024

Data scientist

Men's Health & Reproductive Health Research Centre, Shohadaye Tajrish Hospital · Tehran, Iran

- Multimodal deep learning for semen analysis and drug-effectiveness prediction from clinical data and imaging.

INDUSTRY EXPERIENCE

2023 – 2025

Data scientist

HiWEB · Tehran, Iran

- Churn, next-purchase and purchase-value models for marketing; ASR and LLM-based quality control for Persian call-centre audio; dashboards in Flask and Plotly Dash.

2021

Data scientist

YecomSoft · Tehran, Iran

- Persian NLP and text-to-speech for audiobook production.

2020 □ **Django web developer (intern)**

MAPSA · Tehran, Iran

- Backend and REST APIs for an online retail platform.

PUBLICATIONS

[9] Bioprinting-Enabled Customization of Biohybrid Actuators: Scaling Dimensions and Functional Complexity

Pajouhande M, Celikkin N, **Rafieyan S**, et al.

Gels · 2026 · Review · doi:10.3390/gels12100923

[8] MLATE V3: An Open-Source Multi-Tissue AI Framework for Data-Driven Optimization of 3D-Printed and Bioprinted Scaffolds

Rafieyan S, Partovi Nasr M, Ansari E, Kheradvar Kolour A, Banimohamad-Shotorbani B, Vasheghani-Farahani E

Biofabrication · 2026 · doi:10.1088/1758-5090/aeb162 ONLINE AHEAD OF PRINT

[7] Data-driven insights for enhanced cellulose conversion to 5-hydroxymethylfurfural using machine learning

Qiao Y, Kargaran E, Ji H, Madadi M, **Rafieyan S**, Liu D

Bioresource Technology · 2025 · doi:10.1016/j.biortech.2025.132582

[6] A practical machine learning approach for predicting the quality of 3D (bio)printed scaffolds

Rafieyan S, Ansari E, Vasheghani-Farahani E

Biofabrication · 2024 · doi:10.1088/1758-5090/ad6374

[5] Acetone-butanol-ethanol fermentation products recovery: Challenges and opportunities

Rafieyan S, Boojari MA, Setayeshnia A, et al.

Chemical Engineering Research and Design · 2024 · Review · doi:10.1016/j.cherd.2024.04.021

[4] MLATE: Machine learning for predicting cell behavior on cardiac tissue engineering scaffolds

Rafieyan S, Vasheghani-Farahani E, Baheiraei N, Keshavarz H

Computers in Biology and Medicine · 2023 · doi:10.1016/j.compbiomed.2023.106804

[3] **A Review of Recent Advances in Natural Polymer-Based Scaffolds for Musculoskeletal Tissue Engineering**

Fan J, Abedi-Dorcheh K, Sadat Vaziri A, Kazemi-Aghdam F, **Rafieyan S**, et al.

Polymers · 2022 · Review · doi:10.3390/polym14102097

[2] **Predicting cell behavior on cardiac tissue engineering scaffolds using machine learning algorithms**

Rafieyan S

MSc thesis, Tarbiat Modares University, Tehran · 2022

[1] **Process Simulation of Biodiesel Production from Soybean Oil with Aspen HYSYS: A Comparative Study of two different processes**

Rafieyan S, Bayat A

11th International Chemical Engineering Congress & Exhibition (IChEC 2020), Fouman, Iran · 2020 · Conference paper

SOFTWARE AND DATA

MLATE (Machine Learning Applications in Tissue Engineering). An open-source framework that uses literature-derived data and machine learning to predict scaffold printability and cell response, helping researchers prioritise candidate formulations for experimental testing.

TissueGPT (doi:10.57967/hf/8917). A language model adapted to tissue-engineering literature for exploring domain-specific text generation and research assistance.

PGG (Porous Geometry Generation). A public tool for generating porous geometries for research and fabrication.

SERVICE

Peer reviewer for IOP Publishing journals (IOP Trusted Reviewer), including Physics in Medicine and Biology, Biomedical Materials, Biomedical Physics & Engineering Express, Engineering Research Express and Progress in Biomedical Engineering; also Results in Surfaces and Interfaces (Elsevier).

SKILLS

Machine learning

Regression, classification, clustering, feature engineering, model selection, Bayesian optimisation, gradient boosting (XGBoost, LightGBM, CatBoost), scikit-learn

Deep learning and NLP

PyTorch, TensorFlow, Hugging Face Transformers, CNNs, RNNs, multimodal learning, LLM fine-tuning, text mining, topic modelling

Programming and tools

Python, NumPy, pandas, Matplotlib, Plotly, Dash, Streamlit, Flask, FastAPI, Docker, GitHub Actions, PostgreSQL, MySQL, SQLite

Laboratory

Cell culture, MTT assay, decellularisation, hydrogel preparation, scaffold fabrication, 3D printing, freeze-drying

LANGUAGES

English · TOEFL iBT 90/120

Persian · Native

ADDITIONAL TRAINING

Neural Networks and Deep Learning · Coursera · 2020

Deep Learning with PyTorch: Image Segmentation · Coursera · 2022

Deep Learning with PyTorch: Object Localization · Coursera · 2022

Data Visualization using Plotly · Coursera · 2022

Introduction to Genomic Technologies · Coursera · 2022

Python for Genomic Data Science · Coursera · 2022

REFERENCES

Available on request.